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QUANTUM DEVICE

Abstract

A quantum device includes a filter having a plurality of notches in a first frequency band lower than a second frequency band, a first qubit that resonates in the first frequency band, and a second qubit that resonates in the second frequency band. The first qubit is input with a signal having a frequency at which the first qubit resonates via the filter to control a state of the first qubit when executing a one-qubit gating, and is input with a signal having a frequency at which the second qubit resonates via the filter to control a state of the second qubit according to a quantum state of the first qubit when executing a two-qubit gating.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2023-149424, filed on Sep. 14, 2023, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The embodiments discussed herein are related to quantum devices.

BACKGROUND

[0003] In a quantum computer using superconducting quantum bits (hereinafter, also simply referred to as a “qubits”), there is a known method of performing a two-qubit gating operation. In order to control the qubit, the qubit is coupled to a waveguide, and a microwave pulse is irradiated from the waveguide. The stronger the coupling between the qubit and the waveguide is, the shorter the time (or gating time) required to perform the qubit gating operation becomes. On the other hand, the stronger the coupling between the qubit and the waveguide is, the shorter the quantum lifetime (or quantum coherence) of the qubit becomes. That is, increasing the quantum lifetime of the quantum bit and reducing the gating time are in a mutually contradictory relationship, as explained in International Publication Pamphlet No. WO 2020/105732, for example.

[0004] In a two-qubit gating (also referred to as a cross resonance gating), a signal having a frequency at which a target qubit resonates is input from the waveguide to a control qubit, such that a state of the target qubit changes according to a state of the control qubit.

[0005] In the case of the two-qubit gating, the gating can be speeded up (that is, the gating time can be shortened) by increasing an intensity of the signal input to the control qubit or increasing a coupling efficiency between the waveguide and the qubit. However, if the coupling between the control qubit and the input waveguide is too strong, a radiation loss from the control qubit to the input waveguide increases, and a coherence time (or data retention time) of the control qubit is shortened.

SUMMARY

[0006] Accordingly, it is an object in one aspect of the embodiments to provide a quantum device capable of securing a quantum gating speed and a coherence time.

[0007] According to one aspect of the embodiments, a quantum device includes a filter having a plurality of notches in a first frequency band lower than a second frequency band; a first qubit that resonates in the first frequency band; and a second qubit that resonates in the second frequency band, wherein the first qubit is input with a signal having a frequency at which the first qubit resonates via the filter to control a state of the first qubit when executing a one-qubit gating, and is input with a signal having a frequency at which the second qubit resonates via the filter to control a state of the second qubit according to a quantum state of the first qubit when executing a two-qubit gating.

[0008] The object and advantages of the invention will be realized and attained by means of the elements and combinations particularly pointed out in the claims.

[0009] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive of the invention, as claimed.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. 1 is a diagram illustrating a connection example of qubits;

[0011] FIG. 2 is a diagram for explaining a frequency dependence of a coupling efficiency between

a waveguide and a control qubit;

[0012] FIG. **3** is a diagram for explaining a quantum device according to one embodiment;

[0013] FIG. **4** is a diagram illustrating an example of a frequency characteristic of the coupling efficiency of a filter;

[0014] FIG. **5** is a perspective view illustrating a configuration example of the quantum device according to a first embodiment;

[0015] FIG. **6** is an equivalent circuit diagram of the quantum device according to the first embodiment;

[0016] FIG. **7** is a plan view illustrating a configuration example of the quantum device according to a second embodiment;

[0017] FIG. **8** is an equivalent circuit diagram of the quantum device according to the second embodiment; and

[0018] FIG. **9** is a diagram for explaining a frequency characteristic of the coupling efficiency of the filter and resonance frequencies of qubits.

DESCRIPTION OF EMBODIMENTS

[0019] Preferred embodiments of the present invention will be described with reference to the accompanying drawings.

[0020] A quantum device according to one embodiment includes a filter for correcting a level difference between signals required for a one-qubit gating and a two-qubit gating, which are input to each qubit, at each resonance frequency of a control qubit that resonates in a low frequency band and a target qubit that resonates in a high frequency band, in order to implement a quantum logic gate of a quantum computer. The quantum computer includes a one-qubit logic gate such as a Pauli gate or the like, and a two-qubit logic gate such as a controlled Pauli-X gate (or CNOT gate) or the like, and performs a computation using these qubit logic gates. The CNOT gate is used for generating a state of quantum entanglement or the like.

[0021] A cross resonance gating is an example of a configuration for inducing the quantum entanglement. The cross resonance gating has a control qubit adjacent to an input end, and a target qubit to be coupled to the control qubit. The control qubit and the target qubit have mutually different resonance frequencies.

[0022] FIG. **1** is a diagram illustrating a connection example of qubits. A control signal S for driving a two-qubit gating (or cross resonance gating) in which a control qubit **10** and a target qubit **20** are coupled, is input to the control qubit **10**.

[0023] In the cross resonance gating, the control signal S having a frequency identical to a resonance frequency (or resonant frequency) of the target qubit **20** is input to the control qubit **10**, in order to induce the quantum entanglement between the control quantum bit **10** and the target qubit **20** that are mutually adjacent. Hence, a state of the target qubit **20** changes according to a quantum state of the control qubit **10**. The control signal S is, for example, a microwave.

[0024] In the case of the cross resonance gating in which the control qubit **10** resonating in a low frequency band and the target qubit **20** resonating in a high frequency band are coupled, the gating can be speeded up (the gating time can be shortened) by increasing an intensity of the control signal S input from a waveguide **31** to the control qubit **10** or increasing a coupling efficiency of the control signal S from the waveguide **31** to the control qubit **10**. However, if the coupling intensity between the waveguide **31** and the control qubit **10** is too large, a coherence time (or data retention time) of the control qubit **10** is shortened.

[0025] Hence, the quantum device according to the present embodiment includes a filter structure in which a coupling efficiency γ between the waveguide **31** and the control qubit **10** has a frequency dependence illustrated in FIG. **2**. The coupling efficiency γ represents a ratio of an intensity of the control signal S input from the waveguide **31** to the control qubit **10** with respect to an intensity of the control signal S input to the waveguide **31**. The intensity of the control signal S is represented by a magnitude of a power of the control signal S, for example. The filter structure

according to the present embodiment has a coupling characteristic such that the coupling efficiency γ in a second frequency band including a frequency (resonant frequency $\omega_{\text{sub.2}}$) at which the target qubit **20** resonates is high compared to the coupling efficiency γ in a first frequency band including a frequency (resonant frequency $\omega_{\text{sub.1}}$) at which the control qubit **10** resonates.

[0026] Because the coupling efficiency γ is high at the resonant frequency $\omega_{\text{sub.2}}$, the control signal S having a relatively high intensity at the resonant frequency $\omega_{\text{sub.2}}$ is input to the control qubit **10**, and thus, it is possible to speed up the two-qubit gating between the control qubit **10** and the target qubit **20**. On the other hand, because the coupling efficiency γ is low at the resonant frequency $\omega_{\text{sub.1}}$, it is possible to reduce a radiation loss from the control qubit **10** to the waveguide **31** at the resonant frequency $\omega_{\text{sub.1}}$, and reduce shortening of the coherence time of the control qubit **10**. Accordingly, it is possible to provide a two-qubit gating capable of securing the quantum gating speed and the coherence time. For example, by adopting a filter

[0027] structure having a high coupling efficiency γ at the resonant frequency $\omega_{\text{sub.2}}$, the quantum gating speed can be secured compared to a case where such a filter structure is not adopted, even if the power of the control signal S input to the waveguide **31** at the resonant frequency $\omega_{\text{sub.2}}$ is reduced. By reducing the power of the control signal S input to the waveguide **31** at the resonant frequency $\omega_{\text{sub.2}}$, it is possible to achieve a reduced power consumption.

[0028] For example, the power of the control signal S at the resonant frequency $\omega_{\text{sub.2}}$ can be reduced to the power of the control signal S at the resonant frequency $\omega_{\text{sub.1}}$. It is advantageous for digital generation of the control signal S when the power of the control signal S at the resonant frequency $\omega_{\text{sub.2}}$ and the power of the control signal S at the resonant frequency $\omega_{\text{sub.1}}$ become equal. For example, if signals having a large level difference are created using a single digital-to-analog converter (DAC), a dynamic range of the signals becomes narrow, and a quality of the signals may deteriorate. On the other hand, if such signals having the large level difference are created using a plurality of DACs, a number of required components or parts increases, and cause an increase in the size and cost of the device for generating the control signal S . In contrast, when the power of the control signal S at the resonant frequency $\omega_{\text{sub.2}}$ and the power of the control signal S at the resonant frequency $\omega_{\text{sub.1}}$ become equal, it is possible to reduce the deterioration of the signal quality caused by an insufficient dynamic range, even when the control signal S is generated using a single DAC.

[0029] A quantum device **100** according to the present embodiment illustrated in FIG. **3** includes a highpass filter **40** in a stage preceding the control qubit **10**. The highpass filter has a coupling characteristic in which the coupling efficiency γ in a second frequency band including the resonant frequency $\omega_{\text{sub.2}}$ is higher than the coupling efficiency γ in a first frequency band including the resonant frequency $\omega_{\text{sub.1}}$.

[0030] FIG. **3** is a diagram illustrating a configuration example of the quantum device according to the present embodiment. The quantum device **100** illustrated in FIG. **3** includes the highpass filter **40** that transmits the control signal S in the first frequency band with the coupling efficiency γ lower than that in the second frequency band, the control qubit **10** that resonates in the first frequency band, and the target qubit **20** that resonates in the second frequency band. The quantum device **100** is a cross resonance gating in which the control qubit **10** and the target qubit **20** are coupled via a transmission line **30**, for example.

[0031] The highpass filter **40** is an example of a filter that transmits a signal in a first frequency band with a coupling efficiency γ lower than that in a second frequency band. The control qubit **10** is an example of a first qubit that resonates in the first frequency band. The target qubit **20** is an example of a second qubit that resonates in the second frequency band.

[0032] When implementing a one-qubit gating, a signal having a frequency at which the control qubit **10** resonates is input to the control qubit **10** via the highpass filter **40**, thereby controlling a quantum state of the control qubit **10**. When implementing a two-qubit gating, the control signal S having a frequency at which the target qubit **20** resonates is input to the control qubit **10** via the

highpass filter **40**. As a result, a state of the target qubit **20** changes according to the quantum state of the control qubit **10**.

[0033] The highpass filter **40** included in the quantum device **100** according to the present embodiment has a plurality of notches in a low frequency band (the first frequency band in which the coupling efficiency γ is reduced more than in the second frequency band) including the resonance frequency of the control qubit **10**. Hence, it is possible to secure the first frequency band in which the coupling efficiency γ is reduced more than in the second frequency band to have a width that is wide to a certain extent. Because the wide width of the first frequency band can be secured, a coupling efficiency characteristic of the highpass filter **40** in the first frequency band can be made to approximate a flat characteristic. As a result, even when the frequency at which the control qubit **10** resonates is shifted due to manufacturing variations or the like, it is possible to stabilize the operation of the quantum device **100** (cross resonance gating).

[0034] FIG. **4** is a diagram illustrating an example of a frequency characteristic of a coupling efficiency of a filter. The highpass filter **40** has a coupling efficiency characteristic that transmits the control signal **S** in a first frequency band **W1** with a coupling coefficient γ lower than that in a second frequency band **W2**. The highpass filter **40** has a plurality of notches (in this example, a first notch **51** and a second notch **52**) in the first frequency band **W1** including a resonant frequency f_{q1} of the control qubit **10**.

[0035] The first notch **51** is a frequency characteristic point at which a control signal **S1** is minimized at a first notch frequency f_{n1} included in the first frequency band **W1**. The second notch **52** is a frequency characteristic point at which the control signal **S1** is minimized at a second notch frequency f_{n2} included in the first frequency band **W1**. The second notch frequency f_{n2} is higher than the first notch frequency f_{n1} .

[0036] The highpass filter **40** has a frequency characteristic **A1** having the first notch **51** and the second notch **52** in the first frequency band **W1**. Thus, the width of the first frequency band **W1** in which the control signal **S** input to the control qubit **10** is transmitted with the coupling coefficient γ lower than that of the second frequency band **W2** can be widened compared to the case of a frequency characteristic **A2** having one notch **53** having a notch frequency f_{n3} in a low frequency band. Because the width of the first frequency band **W1** in which the control signal **S** is transmitted with the coupling coefficient γ lower than that of the second frequency band **W2** can be widened, the coupling coefficient characteristic of the highpass filter **40** in the first frequency band **W1** can be made to approximate a flat characteristic. As a result, even when the frequency at which the control qubit **10** resonates is shifted due to the manufacturing variations or the like, it is possible to stabilize the operation of the quantum device **100** (cross resonance gating).

[0037] In the example illustrated in FIG. **4**, the highpass filter **40** has the first notch **51** and the second notch **52** such that a resonance frequency f_{q1} of the control qubit **10** is interposed between the first notch frequency f_{n1} and the second notch frequency f_{n2} . In this case, the control qubit **10** resonates in a frequency band interposed between the first notch frequency f_{n1} and the second notch frequency f_{n2} . Hence, by providing the notches above and below the resonance frequency f_{q1} of the control qubit **10**, it is possible to prevent the coupling efficiency γ at the resonance frequency f_{q1} from becoming excessively reduced.

[0038] FIG. **4** illustrates an example of the frequency characteristic **A1** in which the highpass filter **40** has two notches, including the first notch **51** and the second notch **52**, in the first frequency band **W1**. However, the highpass filter **40** may have three or more notches in the first frequency band **W1**. Thus, it is possible to further widen the width of the first frequency band **W1** in which the coupling coefficient γ is reduced more than in the second frequency band **W2**.

[0039] The highpass filter **40** may have a frequency characteristic having one or more resonance points that resonate between the resonance frequency f_{q1} of the control qubit **10** and the resonance frequency f_{q2} of the target qubit **20**. The example of the frequency characteristic **A1** illustrated in FIG. **4** has a first resonance point **54** having a filter resonance frequency f_{r1} and a second

resonance point **55** having a filter resonance frequency **fr2**, between the resonance frequency **fq1** and the resonance frequency **fq2**. The filter resonance frequency **fr1** is higher than the second notch frequency **fn2**. The filter resonance frequency **fr2** is higher than the filter resonance frequency **fr1**. The frequency characteristic **A2** has one resonance point **56** having a filter resonance frequency **fr3** between the resonance frequency **fq1** and the resonance frequency **fq2**.

[0040] Because the highpass filter **40** has the frequency characteristic **A1** that resonates between the resonance frequency **fq1** and the resonance frequency **fq2**, it is possible to increase a coupling efficiency difference **D** compared to the case where the highpass filter **40** has a frequency characteristic that does not resonate between the resonance frequency **fq1** and the resonance frequency **fq2**. The coupling efficiency difference **D** represents a difference between the coupling efficiency γ of the highpass filter **40** in the first frequency band **W1** and the coupling efficiency γ of the highpass filter **40** in the second frequency band **W2**. By increasing the coupling efficiency difference **D**, it is possible to reduce the intensity of the control signal **S** in the first frequency band **W1**, even if the intensity of the control signal **S** in the second frequency band **W2** is increased. That is, it is possible to simultaneously increase the quantum gating speed and reduce a decrease in the coherence time.

[0041] The highpass filter **40** may have a coupling efficiency characteristic that transmits the control signal **S** in the first frequency band **W1** with a coupling coefficient γ lower than that in the second frequency band **W2**, by utilizing Fano resonance. By utilizing the Fano resonance, it is possible to easily obtain a frequency characteristic **A1** that can secure a large coupling efficiency difference **D**.

[0042] As illustrated in FIG. 3, the highpass filter **40** may include a first filter **41** forming the first notch **51**, and a second filter **42** forming the second notch **52**. Because each notch is formed by a separate filter, it is possible to improve the accuracy of the first notch frequency **fn1** and the second notch frequency **fn2**. For example, the first filter **41** can implement a frequency characteristic having the first notch **51** and the first resonance point **54** by utilizing the Fano resonance. The second filter **42** can implement a frequency characteristic having the second notch **52** and the second resonance point **55** by utilizing the Fano resonance. The frequency characteristic **A1** illustrated in FIG. 4 can be achieved by a combination of the frequency characteristic of the first filter **41** and the frequency characteristic of the second filter **42**.

[0043] FIG. 5 is a diagram illustrating a configuration example of the quantum device according to a first embodiment. A quantum device **101** illustrated in FIG. 5 is an example of the quantum device **100** described above. The quantum device **101** illustrated in FIG. 5 includes a substrate **70** having a first layer **71** facing a first direction, and a second layer **72** facing a second direction opposite to the first direction. One of the first layer **71** and the second layer **72** is a front surface layer of the substrate **70**, and the other of the first layer **71** and the second layer **72** is a rear surface layer of the substrate **70**. The substrate **70** is a dielectric substrate on which a predetermined conductor pattern is formed.

[0044] A terminal **43** to which the control signal **S** is input is provided on the first layer **71**. A terminal **43** is electrically connected to a coaxial line **60** that transmits the control signal **S**. The terminal **43** is an electrode, such as a planar terminal pad or the like, for example. The coaxial line **60** is an example of a transmission line, such as a coaxial cable or the like, for example. The coaxial line **60** is an example of the waveguide **31** described above. The control signal **S** input to the terminal **43** via the coaxial line **60** passes through an inner layer of the substrate **70**, and reaches the control qubit **10** provided on the second layer **72**. Although not illustrated in FIG. 5, the target qubit **20** coupled to the control qubit **10** is provided on the second layer **72**, for example.

[0045] Because the quantum device **101** has the configuration in which the control signal **S** is transmitted from the first layer **71** to the second layer **72**, a length of a transmission line for transmitting the control signal **S** from the terminal **43** to the control qubit **10** is relatively short (approximately equal to a thickness of the substrate **70**). For this reason, according to conventional

techniques, it is difficult to provide a highpass filter having a coupling efficiency characteristic like the frequency characteristic A1 described above on the transmission line from the terminal 43 to the control qubit 10, due to space constraints.

[0046] Accordingly, in the quantum device 101 according to the first embodiment, the highpass filter 40 (more specifically, the first filter 41 and the second filter 42), which utilizes an interference of the Fano resonance for the coupling efficiency characteristic, is provided on the first layer 71. Hence, even in the quantum device 101 having a configuration in which it is difficult to sufficiently secure the length of the transmission line for transmitting the control signal S from the terminal 43 to the control qubit 10, it is easy to form the highpass filter 40 having the frequency characteristic A1 described above.

[0047] The first filter 41 and the second filter 42 are provided around the terminal 43. Thus, it is easy to form the highpass filter 40 having the frequency characteristic A1 described above that transmits the control signal S input to the control qubit 10 from the terminal 43 with the low coupling coefficient γ . In this example, the second filter 42 is formed so as to oppose the first filter 41 with the terminal 43 interposed therebetween.

[0048] The first filter 41 includes a first resonator 44 adjacent to the terminal 43, and the second filter 42 includes a second resonator 45 adjacent to the terminal 43. The first resonator 44 is a high-frequency circuit that forms the first resonance point 54 of the frequency characteristic A1, and the second resonator 45 is a high-frequency circuit that forms the second resonance point 55 of the frequency characteristic A1. The second resonator 45 is formed so as to oppose the first resonator 44 with the terminal 43 interposed therebetween. The first resonator 44 is disposed between the terminal 43 and a ground 73. The second resonator 45 is disposed between the terminal 43 and the ground 73. The ground 73 is formed on the first layer 71.

[0049] The first resonator 44 includes a first conductor pattern 46 formed on the first layer 71. The second resonator 45 includes a second conductor pattern 47 formed on the first layer 71. The first conductor pattern 46 and the second conductor pattern 47 are linear conductors formed in a fan-shaped area illustrated in FIG. 4 and folded back a plurality of times, for example.

[0050] The control qubit 10 includes a Josephson junction 11 that does not overlap the first resonator 44 and the second resonator 45 in a plan view of the substrate 70. Thus, effects of crosstalk of the first resonator 44 and the second resonator 45 on the Josephson junction 11 can be reduced, and thus, the coherence time (an average time for which the qubit maintains an excited state of "1") of the control qubit 10 can be maintained.

[0051] FIG. 6 is an equivalent circuit diagram of the quantum device according to the first embodiment illustrated in FIG. 5. The substrate 70 has a substrate capacitance $C_{\text{sub.F}}$ between the terminal 43 and the control qubit 10. The substrate capacitance $C_{\text{sub.F}}$ is a coupling capacitance between the terminal 43 and the control qubit 10, and is included in the first filter 41 and the second filter 42.

[0052] The first filter 41 includes a first capacitance $C_{\text{sub.k}}$ between the terminal 43 and the first conductor pattern 46 of the first resonator 44, and a second capacitance $C_{\text{sub.q}}$ between the first conductor pattern 46 of the first resonator 44 and the control qubit 10. That is, the first conductor pattern 46 forms the first capacitance $C_{\text{sub.k}}$ between the first conductor pattern 46 and the terminal 43, and forms the second capacitance $C_{\text{sub.q}}$ between the first conductor pattern 46 and the control qubit 10. The first notch 51 is formed by a destructive interference (Fano resonance) between a signal transmitted to the control qubit 10 via the first resonator 44 by the first capacitance $C_{\text{sub.k}}$ and the second capacitance $C_{\text{sub.q}}$ and a signal transmitted directly to the control qubit 10 from the waveguide via the substrate capacitance $C_{\text{sub.F}}$.

[0053] The second filter 42 includes a third capacitance $C_{\text{sub.k2}}$ between the terminal 43 and the second conductor pattern 47 of the second resonator 45, and a fourth capacitance $C_{\text{sub.q2}}$ between the second conductor pattern 47 of the second resonator 45 and the control qubit 10. That is, the second conductor pattern 47 forms the third capacitance $C_{\text{sub.k2}}$ between the second conductor

pattern **47** and the terminal **43**, and forms the fourth capacitance $C_{\text{sub.q2}}$ between second conductor pattern **47** and the control qubit **10**. The second notch **52** is formed by a destructive interference between a signal transmitted to the control qubit **10** via the second resonator **45** by the third capacitance $C_{\text{sub.k2}}$ and the fourth capacitance $C_{\text{sub.q2}}$ and a signal transmitted directly to the control qubit **10** from the waveguide via the substrate capacitance $C_{\text{sub.F}}$.

[0054] The first resonator **44** includes a fifth capacitance $C_{\text{sub.R}}$ between the first conductor pattern **46** and the ground **73**, and a first inductor $L_{\text{sub.R}}$ between the first conductor pattern **46** and the ground **73**. The first resonator **44** is an LC resonator including the fifth capacitance $C_{\text{sub.R}}$ and the first inductor $L_{\text{sub.R}}$. The first resonance point **54** is formed by the first resonator **44**.

[0055] The second resonator **45** includes a sixth capacitance $C_{\text{sub.R2}}$ between the second conductor pattern **47** and the ground **73**, and a second inductor $L_{\text{sub.R2}}$ between the second conductor pattern **47** and the ground **73**. The second resonator **45** is an LC resonator including the sixth capacitance $C_{\text{sub.R2}}$ and the second inductor $L_{\text{sub.R2}}$. The second resonance point **55** is formed by the second resonator **45**.

[0056] FIG. **7** is a plan view illustrating a configuration example of the quantum device according to a second embodiment. A quantum device **102** illustrated in FIG. **7** is an example of the quantum device **100** described above. The quantum device **102** illustrated in FIG. **7** includes a substrate **70** having an input line **75** to which the control signal S is input and an output line **76** to which the control qubit **10** is connected. The quantum device **102** includes a highpass filter **40** including a plurality of filters (in the example, a first filter **41** and a second filter **42**) connected in series between the input line **75** and the output line **76**. The first filter **41** includes a first resonator **44** adjacent to the input line **75**, and the second filter **42** includes a second resonator **45** adjacent to the output line **76**. The first resonator **44** is a high-frequency circuit that forms a first resonance point **54** of the frequency characteristic **A1**, and the second resonator **45** is a high-frequency circuit that forms a second resonance point **55** of the frequency characteristic **A1**. The second resonator **45** is formed so as to face the first resonator **44** with a capacitive coupling interposed therebetween. The first resonator **44** is disposed between the input line **75** and the second resonator **45** so as to be surrounded by a ground **73**. The second resonator **45** is disposed between the first resonator **44** and the output line **76** so as to be surrounded by the ground **73**. The ground **73** is formed on a first layer **71**.

[0057] The first resonator **44** includes a first conductor pattern **46** formed on the first layer **71**. The second resonator **45** includes a second conductor pattern **47** formed on the first layer **71**. The first conductor pattern **46** and the second conductor pattern **47** are linear conductors folded back a plurality of times.

[0058] FIG. **8** is an equivalent circuit diagram of the quantum device of the second embodiment illustrated in FIG. **7**. The quantum device **102** includes a capacitance $C_{\text{sub.RE1}}$ between the input line **75** and the first filter **41**, and a capacitance $C_{\text{sub.RE2}}$ between the first filter **41** and the second filter **42**. The quantum device **102** further includes a capacitance $C_{\text{sub.RE3}}$ between the input line **75** and the output line **76** or the control qubit **10**, and a capacitance $C_{\text{sub.QR}}$ between the second filter **42** and the output line **76** or the control qubit **10**. The plurality of notches including the first notch **51** and the second notch **52** are formed by a destructive interference between a signal transmitted from the input line **75** to the output line **76** via the first resonator **44** and the second resonator **45** by the capacitance $C_{\text{sub.RE1}}$, the capacitance $C_{\text{sub.RE2}}$, and the capacitance $C_{\text{sub.QR}}$ and a signal transmitted directly to the output line **76** from the input line **75** by the capacitance $C_{\text{sub.RE3}}$.

[0059] The first resonator **44** includes a capacitance $C_{\text{sub.R}}$ between the first conductor pattern **46** and the ground **73**, and a first inductor $L_{\text{sub.R}}$ between the first conductor pattern **46** and the ground **73**. The first resonator **44** is an LC resonator including the capacitance $C_{\text{sub.R}}$ and the first inductor $L_{\text{sub.R}}$. The first resonance point **54** is formed by the first resonator **44**.

[0060] The second resonator **45** includes a capacitance $C_{\text{sub.R2}}$ between the second conductor

pattern **47** and the ground **73**, and a second inductor $L_{\text{sub.R2}}$ between the second conductor pattern **47** and the ground **73**. The second resonator **45** is an LC resonator including the capacitance $C_{\text{sub.R2}}$ and the second inductor $L_{\text{sub.R2}}$. The second resonance point **55** is formed by the second resonator **45**.

[0061] FIG. **9** is a diagram for explaining the frequency characteristic of the coupling efficiency of the filter and the resonance frequency of each qubit. The target qubit **20** may include a plurality of target qubits (in this example, four target qubits **20a**, **20b**, **20c**, and **20d**) that resonate at mutually different frequencies in the second frequency band **W2**.

[0062] The control qubit **10** has a resonance frequency f_{q1} within the first frequency band **W1**. By providing a plurality of notches in the first frequency band **W1**, the control signal **S** input to the control qubit **10** is transmitted in the low frequency band including the resonance frequency f_{q1} with a coupling coefficient γ lower than that in the second frequency band **W2**.

[0063] On the other hand, the plurality of target qubits **20a**, **20b**, **20c**, and **20d** have resonance frequencies f_{q2a} , f_{q2b} , f_{q2c} , and f_{q2d} within the second frequency band **W2**. The highpass filter **40** has a coupling coefficient γ in the first frequency band **W1** lower than a coupling coefficient γ in the second frequency band **W2**. The plurality of target qubits **20a**, **20b**, **20c**, and **20d** resonate at frequencies included in a high frequency band having a flat characteristic in the frequency characteristic of the highpass filter **40**.

[0064] According to the present disclosure, the quantum gating speed and the coherence time can be secured.

[0065] Although the embodiments are numbered with, for example, “first,” or “second,” the ordinal numbers do not imply priorities of the embodiments. Many other variations and modifications will be apparent to those skilled in the art.

[0066] All examples and conditional language recited herein are intended for pedagogical purposes to aid the reader in understanding the invention and the concepts contributed by the inventor to furthering the art, and are to be construed as being without limitation to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to a showing of the superiority and inferiority of the invention. Although the embodiments of the present invention have been described in detail, it should be understood that the various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

[0067] Various aspects of the subject-matter described herein may be set out non-exhaustively in the following numbered clauses:

Clause 1

[0068] A quantum device comprising: [0069] a filter having a plurality of notches in a first frequency band lower than a second frequency band; [0070] a first qubit that resonates in the first frequency band; and [0071] a second qubit that resonates in the second frequency band, [0072] wherein the first qubit is input with a signal having a frequency at which the first qubit resonates via the filter to control a state of the first qubit when executing a one-qubit gating, and is input with a signal having a frequency at which the second qubit resonates via the filter to control a state of the second qubit according to a quantum state of the first qubit when executing a two-qubit gating.

Clause 2

[0073] The quantum device according to clause 1, wherein the filter resonates between a resonant frequency of the first qubit and a resonant frequency of the second qubit.

Clause 3

[0074] The quantum device according to clause 1 or 2, wherein the plurality of notches includes a first notch having a first notch frequency, and a second notch having a second notch frequency higher than the first notch frequency and lower than a resonance frequency of the filter.

Clause 4

[0075] The quantum device according to clause 3, wherein the filter includes a first filter forming

the first notch, and a second filter forming the second notch.

Clause 5

[0076] The quantum device according to clause 4, further comprising: [0077] a substrate including a first layer provided with the first filter and the second filter, and a second layer provided with the first qubit.

Clause 6

[0078] The quantum device according to clause 5, wherein: [0079] the first layer is provided with a terminal to which a signal having a frequency at which the first qubit or the second qubit resonates is input, and [0080] the first filter and the second filter are provided around the terminal.

Clause 7

[0081] The quantum device according to clause 6, wherein the second filter opposes the first filter via the terminal interposed between the first filter and the second filter.

Clause 8

[0082] The quantum device according to clause 6 or 7, wherein: [0083] the first filter includes a first resonator in close proximity to the terminal, and [0084] the second filter includes a second resonator in close proximity to the terminal.

Clause 9

[0085] The quantum device according to clause 8, wherein the first qubit includes a Josephson junction that does not overlap the first resonator and the second resonator in a plan view of the substrate.

Clause 10

[0086] The quantum device according to clause 8 or 9, wherein: [0087] the first filter includes a first capacitance between the terminal and the first resonator, and a second capacitance between the first resonator and the first qubit, and [0088] the second filter includes a third capacitance between the terminal and the second resonator, and a fourth capacitance between the second resonator and the first qubit.

Clause 11

[0089] The quantum device according to any one of clauses 8 to 10, wherein: [0090] each of the first filter and the second filter includes a coupling capacitance between the terminal and the first qubit, and [0091] an interference between a signal via the first resonator and a signal via the coupling capacitance forms the first notch, and an interference between a signal via the second resonator and the signal via the coupling capacitance forms the second notch.

Clause 12

[0092] The quantum device according to clause 10, wherein: [0093] the first resonator includes a first conductor pattern that forms the first capacitance and the second capacitance, a fifth capacitance between the first conductor pattern and a ground, and a first inductor between the first conductor pattern and the ground, and [0094] the second resonator includes a second conductor pattern that forms the third capacitance and the fourth capacitance, a sixth capacitance between the second conductor pattern and the ground, and a second inductor between the second conductor pattern and the ground.

Clause 13

[0095] The quantum device according to any one of clauses 1 to 12, wherein the first qubit resonates in a band between the plurality of notches.

Clause 14

[0096] The quantum device according to any one of clauses 1 to 13, wherein the second qubit includes a plurality of qubits that resonate at frequencies different from one another in the second frequency band.

Clause 15

[0097] The quantum device according to any one of clauses 1 to 14, wherein the filter utilizes Fano resonance.

Claims

1. A quantum device comprising: a filter having a plurality of notches in a first frequency band lower than a second frequency band; a first qubit that resonates in the first frequency band; and a second qubit that resonates in the second frequency band, wherein the first qubit is input with a signal having a frequency at which the first qubit resonates via the filter to control a state of the first qubit when executing a one-qubit gating, and is input with a signal having a frequency at which the second qubit resonates via the filter to control a state of the second qubit according to a quantum state of the first qubit when executing a two-qubit gating.
2. The quantum device as claimed in claim 1, wherein the filter resonates between a resonant frequency of the first qubit and a resonant frequency of the second qubit.
3. The quantum device as claimed in claim 1, wherein the plurality of notches includes a first notch having a first notch frequency, and a second notch having a second notch frequency higher than the first notch frequency and lower than a resonance frequency of the filter.
4. The quantum device as claimed in claim 3, wherein the filter includes a first filter forming the first notch, and a second filter forming the second notch.
5. The quantum device as claimed in claim 4, further comprising: a substrate including a first layer provided with the first filter and the second filter, and a second layer provided with the first qubit.
6. The quantum device as claimed in claim 5, wherein: the first layer is provided with a terminal to which a signal having a frequency at which the first qubit or the second qubit resonates is input, and the first filter and the second filter are provided around the terminal.
7. The quantum device as claimed in claim 6, wherein the second filter opposes the first filter via the terminal interposed between the first filter and the second filter.
8. The quantum device as claimed in claim 6, wherein: the first filter includes a first resonator in close proximity to the terminal, and the second filter includes a second resonator in close proximity to the terminal.
9. The quantum device as claimed in claim 8, wherein the first qubit includes a Josephson junction that does not overlap the first resonator and the second resonator in a plan view of the substrate.
10. The quantum device as claimed in claim 8, wherein: the first filter includes a first capacitance between the terminal and the first resonator, and a second capacitance between the first resonator and the first qubit, and the second filter includes a third capacitance between the terminal and the second resonator, and a fourth capacitance between the second resonator and the first qubit.
11. The quantum device as claimed in claim 8, wherein: each of the first filter and the second filter includes a coupling capacitance between the terminal and the first qubit, and an interference between a signal via the first resonator and a signal via the coupling capacitance forms the first notch, and an interference between a signal via the second resonator and the signal via the coupling capacitance forms the second notch.
12. The quantum device as claimed in claim 10, wherein: the first resonator includes a first conductor pattern that forms the first capacitance and the second capacitance, a fifth capacitance between the first conductor pattern and a ground, and a first inductor between the first conductor pattern and the ground, and the second resonator includes a second conductor pattern that forms the third capacitance and the fourth capacitance, a sixth capacitance between the second conductor pattern and the ground, and a second inductor between the second conductor pattern and the ground.
13. The quantum device as claimed in claim 1, wherein the first qubit resonates in a band between the plurality of notches.
14. The quantum device as claimed in claim 1, wherein the second qubit includes a plurality of

qubits that resonate at frequencies different from one another in the second frequency band.

15. The quantum device as claimed in claim 1, wherein the filter utilizes Fano resonance.
